

AI Agent & OpenClaw Research Intern - Elchai Group - Pre-Interview Assessment

Cash Flow AI Agent

Elchai Group – AI Research Intern Pre-Interview Assessment

Github Repo: <https://github.com/9Ahmed6/cash-flow-ai-agent>

Executive Summary

This report evaluates several open-source AI models referenced in the assessment (Kimi 2.6, Qwen 3.6 Plus, MiniMax, and GLM) to determine which is most suitable for a finance-focused AI assistant.

The proposed proof of concept is **Cash Flow AI Agent**, an AI-powered application that assists finance teams by forecasting cash flow, identifying liquidity risks, and generating executive recommendations.

The objective of this evaluation is not to determine the strongest model overall, but to identify the model that best satisfies the business and technical requirements of this specific use case.

1. Project Objective

Design a proof-of-concept AI solution that assists finance professionals by:

- Forecasting future cash flow
- Detecting liquidity risks
- Generating executive summaries
- Supporting treasury decision-making

The solution is intended to help finance teams rather than automate financial decision-making.

2. Evaluation Methodology

Rather than selecting a model based solely on benchmark performance, the models were evaluated using criteria relevant to enterprise finance workflows.

Evaluation criteria:

- Reasoning capability
- Financial data understanding
- Coding capability
- Cost efficiency
- Local deployment suitability
- AI agent compatibility

3. Candidate Models

Kimi 2.6

Strengths

- Excellent long-context reasoning
- Strong coding capabilities
- Designed for complex autonomous workflows
- Supports large-scale multi-agent orchestration
- Competitive with leading proprietary frontier models according to the reference material

Considerations

The strengths of Kimi are particularly valuable for large autonomous software engineering tasks and extensive multi-agent workflows.

For this project, those capabilities exceed the requirements of a finance decision-support assistant.

Qwen 3.6 Plus (Selected)

Strengths

- Excellent reasoning
- Strong coding performance
- Efficient Mixture-of-Experts architecture
- Supports local deployment
- Strong enterprise ecosystem
- Suitable for structured business workflows

Considerations

Qwen provides an excellent balance between reasoning quality, deployment flexibility, and implementation simplicity.

Its capabilities closely match the needs of financial analysis and executive reporting.

MiniMax

Strengths

- Strong conversational abilities
- Good multimodal support
- Fast inference

Considerations

MiniMax appears better suited to conversational AI applications than structured financial analysis.

GLM

Strengths

- Competitive open-source model
- Strong reasoning capabilities
- Growing ecosystem

Considerations

While technically capable, GLM currently has a smaller ecosystem compared to Qwen for this type of enterprise proof of concept.

Summary

Both Kimi and Qwen are strong candidates for enterprise AI applications.

Kimi demonstrates particular strength in long-running autonomous agent workflows, while Qwen provides a balanced combination of reasoning, coding, efficient deployment, and structured analysis.

For the scope of this proof of concept, Qwen aligns more closely with the project's technical and business requirements.

4. Prototype

A proof-of-concept application was developed to demonstrate a practical finance workflow.

The application enables users to:

- Upload financial transaction data
- Calculate daily cash movement
- Forecast future cash positions
- Estimate liquidity risk
- Generate executive recommendations
- Produce a concise management summary

The objective is to demonstrate how an open-source AI model could support finance teams during treasury planning.

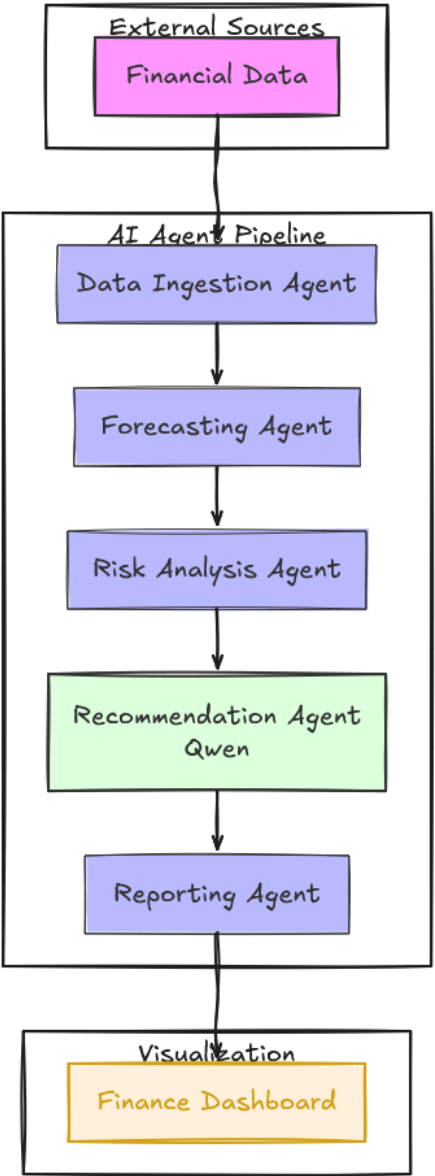
5. AI Agent Framework Integration

While the proof of concept is implemented as a straightforward Python application, a production deployment could benefit from an AI agent framework to orchestrate multiple tasks and integrate enterprise systems.

An AI agent framework provides coordination between specialized components rather than relying on a single model to perform every task. Each agent has a well-defined responsibility and passes structured outputs to the next stage of the workflow.

For the proposed Cash Flow AI Agent, the workflow could be organized as follows:

Agent	Responsibility
Data Ingestion Agent	Collects financial data from CSV files, ERP systems, or accounting platforms and validates the input data.
Forecasting Agent	Calculates projected cash flow for the next 30, 60, and 90 days using the available financial data.
Risk Analysis Agent	Evaluates projected liquidity and identifies potential cash shortages or unusual spending patterns.
Recommendation Agent	Uses Qwen to generate business recommendations based on the forecast and identified risks.
Reporting Agent	Produces an executive summary and prepares a report suitable for finance managers or executives.



Potential Agent Frameworks

Several open-source AI agent frameworks could orchestrate this workflow:

- **OpenClaw** – Suitable for coordinating multiple AI agents and long-running workflows. It aligns with the architecture described in the reference material and could manage interactions between forecasting, risk analysis, and reporting agents.
- **LangGraph** – A graph-based workflow framework that models complex, stateful business processes. It is well suited for enterprise applications where each processing stage depends on previous results.
- **CrewAI** – A role-based multi-agent framework where each agent has a specialized responsibility, making it appropriate for finance workflows that require collaboration between forecasting, analysis, and reporting.

For this proof of concept, a full agent framework was intentionally not implemented in order to keep the solution focused, easy to understand, and reproducible. However, the architecture has been designed so that each module can later be orchestrated by an agent framework without significant changes to the underlying business logic.

6. Test Scenario

Input

A sample dataset containing:

- Daily cash inflows
- Daily cash outflows
- Historical cash balances

Expected Behaviour

The AI should:

- Summarize financial trends
- Forecast future liquidity
- Detect potential liquidity risks
- Generate practical recommendations

Observations

The prototype successfully demonstrated:

- Financial data interpretation
- Forecast generation
- Executive summarization
- Risk explanation
- Action-oriented recommendations

The workflow illustrates how AI can reduce repetitive analysis while keeping finance professionals responsible for final decisions.

7. Elchai Group Business Benefits

The proposed solution offers several potential advantages.

Finance

- Faster cash flow forecasting
- Reduced manual reporting
- Improved financial visibility

Strategy

- Better long-term financial planning
- Improved executive decision support

Operations

- Improved awareness of future cash requirements

IT

- Modular architecture suitable for future enterprise integration

8. Risks and Limitations

Risk	Mitigation
Hallucinated recommendations	Human approval before implementation
Forecast uncertainty	Validate forecasts against historical financial data
Privacy	Deploy within secured enterprise infrastructure
Integration complexity	Begin with a limited proof of concept
Data quality	Validate uploaded financial data
Model reliability	Continuous evaluation and monitoring

The prototype should be viewed as a decision-support system rather than an autonomous financial management solution.

9. Recommendation

Based on this evaluation, **Qwen 3.6 Plus** is recommended for a controlled internal pilot.

The model demonstrates the capabilities required for finance workflow automation while remaining suitable for enterprise deployment and future integration.

The proposed Treasury Decision Support Agent should initially be used to assist finance professionals with forecasting and reporting. Production deployment should include historical validation, governance controls, secure infrastructure, and human oversight for all financial decisions.

If future requirements expand toward highly autonomous, long-running multi-agent workflows, Kimi should be re-evaluated due to its strengths in large-scale agent orchestration.

10. Future Enhancements

Potential future improvements include:

- ERP integration
- Time-series forecasting models
- Scenario analysis
- Monte Carlo cash flow simulation
- Role-based access control
- Audit logging
- Multi-agent orchestration using frameworks such as OpenClaw

11. Conclusion

This assessment demonstrates a structured approach to AI adoption:

1. Research available models.
2. Define business requirements.
3. Evaluate candidate technologies.
4. Build a practical proof of concept.
5. Assess business value.
6. Recommend an adoption strategy.

For this finance use case, Qwen 3.6 Plus provides the strongest balance of reasoning capability, enterprise readiness, and deployment flexibility. The proposed Cash Flow AI Agent illustrates how open-source AI can enhance treasury operations by improving forecasting, reducing manual effort, and supporting informed financial decision-making while preserving human oversight.